Measuring Global Value Chains Integration Using Embodied Labour *

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Abstract

This paper proposes a methodology for analysing global value chain (GVC) integration by measuring vertically integrated embodied labour, rather than relying solely on value-added metrics. We argue that existing GVC indicators suffer from two key shortcomings: they fail to distinguish adequately between GVC and non-GVC trade, and they produce a distorted representation of globalised production as a result of the highly asymmetrical remuneration of production across different geographies. Using input-output analysis with the FIGARO-E3 multi-regional database, we construct sector- and country-level measures of labour embodied in final demand goods. A detailed case study of the German automotive value chain reveals that labour-based metrics depict a significantly more geographically dispersed production structure than value-added measures, which disproportionately emphasise the importance of the most remunerated production steps. Building on our embodied labour measurement framework and on a theoretically clear definition of global value chains, we propose new GVC participation and foreign labour dependency indicators based on embodied labour. These offer a more structurally transparent view of GVCs than their added value counterparts. Our results demonstrate that embodied labour provides a critical complementary lens to value-added analysis, revealing hidden geographies of production and contributing to a more nuanced understanding of the international division of labour.

Keywords: Trade and Labour Market Interactions, InputOutput Analysis, Global Value Chains
JEL Codes: F16, J21, C67

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1 Introduction

Since the mid-1980s, the structure of international trade has undergone a dramatic shift. While the liberalisation and preeminence of trade and capital flows are by no means a novelty in the history of capitalism (Arrighi 2010), what qualitatively distinguishes this "*new wave of globalisation*" is the preeminence of trade in intermediate goods and services, either between subsidiaries of the same firms or through arms-length relations between contractors (Milberg and Winkler 2013). This phenomenon, which was termed "*trade in tasks*" by Grossman and Rossi-Hansberg (2008) and "*the great unbundling*" by Baldwin (2006), is perhaps best explained by illustration. In a seminal work, Linden, Kraemer, and Dedrick (2009) dissect the production chain of one of the early XXIst century's most archetypal commodities: the iPod. The device is made up of about 400 individual components, which are purchased and assembled without ever going through a single Apple-owned manufacturing plant. The entire production is outsourced until the point where the finished products are shipped to Apple distributors. What Apple handles is R&D, design, branding, marketing, and, to some extent, distribution. Yet Apple is the company that extracts the highest margin in the entire operation: USD 80 over the USD 299 price of the iPod model of interest, with the rest of the value being captured by various subcontracts in East Asia, for instance, by Toshiba in Japan for the manufacture of the hard drive.

One framework which international political economy (IPE) scholars have adopted as a way to deal with the increasing level of complexity and integration of international production is that of *global value chains*[1] (GVCs). While early efforts in GVC analysis heavily relied on a case study methodology (Linden, Kraemer, and Dedrick 2009), the increasing preeminence of GVCs in the IPE literature has led to the emergence of more quantitative approaches to the study of GVCs. First, there has been an effort to better disaggregate imports by measuring trade through in value-added rather than the final value (Martins Guilhoto, Webb, and Yamano 2022; OECD 2011). Subsequently, GVC-related macroeconomic indicators have emerged. These indicators have gained significant importance and are used to build stylised facts and assess potential mechanisms for countries to improve income, employment or productivity (OECD 2013; Kümmritz, Taglioni, and Winkler 2017). As has been argued by Carballa Smichowski, Durand, and Knauss (2021), the construction and adoption of these indicators has been problematic as they

[1] Note that "*global value chains*" are also sometimes referred to as "*global commodity chains*" or "*global production networks*". Different authors use definitions which are sometimes overlapping, sometimes equivalent and are inconsistent with each other. We will thus specify in which way we use them *infra*.

are built in ways which disregards the relational understanding of GVCs elaborated by scholarly work (Milberg and Winkler 2013; Durand, Flacher, and Frigant 2018). There are two main aspects which are problematic: first, the most commonly used indicators fail to properly distinguish between GVC and non-GVC trade (Carballa Smichowski, Durand, and Knauss 2021). Second, while the GVC literature insists that a central aspect of GVC production lies in *valuation effects* happening inside the chain (Milberg and Winkler 2013), there are, to our knowledge no GVC-related macroeconomic indicators providing a way to assess the impact of *valuation effects* on GVC integration measurements.

The main contribution of this work, is to propose an alternative approach for the measurement of the integration of a certain national sector in GVCs, that of measuring contribution to GVC production, not in value-added, but in *embodied labour*. We propose an approach for computing vertically integrated labour embodied in production through an input-output analysis method. We compute a decomposition of the vertically integrated labour embodied in production for 45 regional divisons and 76 sectors. Our embodied labour decomposition covers all European countries as well as a diversity of non-European countries of interest for the year 2015. The scope of this decomposition is determined our choice of dataset, in this instance, Figaro-E3 (Cazcarro et al. 2025). We illustrate the relevance of our approach with a case study of Germany's automotive value chain and argue in favour of adopting a synoptic perspective where both value-added and embodied labour perspectives are considered. Furthermore, we build on Carballa Smichowski, Durand, and Knauss (2021)'s proposal of a GVC participation indicator built on a more rigorous theoretical framing by proposing a theoretically consistent, labour-centric indicator to measure country participation and *foreign labour dependency.* We compute our indicators for a sample of 45 countries.

We believe that our approach is relevant for a number of reasons. First, while labour aspects are a mainstay of the scholarly debate around GVC participation, notably through the question of the *race to the bottom* in labour-input cost cutting (Nathan and Sarkar 2011; Milberg and Winkler 2013; Ali 2015), there is a lack of concrete proposals for empirically grounding this discussion. Our methodology addresses this issue. Second, the GVC literature insists that a central aspect of GVC production lies in *valuation effects* inside of GVCs, where prices are driven up or down as a result of either the oligopolistic/oligopsonistic market structure within the chain (Milberg and Winkler 2013; Durand and

Milberg 2020; Perrow 2009) or as a result of practices such as transfer pricing and valuation manipulation operations whereby transnational corporations (TNCs) manipulate prices charged for goods and services exchanged internationally within the firm's subsidiary network in order to minimise their global tax liability (Ali 2015; Heckemeyer and Overesch 2017). This obviously risks introducing biases in the measurement of GVC participation, and there is a definite lack of indicators which are insensitive to price manipulation. Our *embodied labour* approach provides such an indicator. Finally, a growing body of IPE literature discusses the difficulties and risks of misinterpretation when working with aggregated data in a world economy dominated by TNCs and characterised by the preeminence of the routing of financial and trade-flows through offshore financial centres (Ali 2015; Zucman 2013; Gopalan and Rajan 2016). Some authors, such as Ergen, Kohl, and Braun (2023) have insisted on the importance of using disaggregated data as a way of mitigating these challenges. Our methodology yields deeply disaggregated data (both at the sectoral and the regional level) and therefore allows avoiding some of the aforementioned pitfalls.

Applying our embodied labour methodology to the specific case of the German automotive value chain reveals significant disparities between labour-based and value-added-based measurements of global value chain (GVC) integration. We show that value-added metrics disproportionately emphasise domestic economic activity in advanced countries where lead firms are based, such as Germany, in our example. In contrast, embodied labour measurements highlight a more geographically dispersed production structure, with substantial contributions from Eastern Europe (notably Poland and the Czech Republic), China, and the rest of the world. We note a strong negative correlation between labour contribution and value-added per worker, aligning with the hypothesis that production activities are shifted towards low-wage countries.

The rest of the paper is structured as follows. We begin in Section 2 by clarifying the conceptualisation of GVCs which we adopt in this work. We then, in Section 3 introduce our input-output methodology and present our dataset. In Section 4 we showcase the usefulness of our methodology by considering the case study of the German value chain, and follow up in Section 5 by discussing the construction of macroeconomic indicators based on *embodied labour* estimation. We conclude by discussing the relevance of our results, the limitations of our methodology and possible extensions (Section 6).

2 Global Value Chains as a Form of Industrial Organisation

Inquiry into GVCs has emerged as a framework attempting to wrestle with the growing geographical and legal fragmentation of economic space and increasing integration of modern internationalised production. It attempts to provide tools for understanding the organisation of transnational production chains. With roots in world-system theory, the GVCs framework originally emerged as a methodology, one which entailed tracing back across borders which elements (raw material, transformation processes, transportation, etc.) were required to produce a given consumer good (Hopkins and Wallerstein 1977). The paradigm has seen wide adoption from authors with a surprisingly large diversity of theoretical backgrounds: from neoclassical trade theory (Baldwin 2019) to critical geography (Seabrooke and Stausholm 2024) and neo-Marxist theorists (Starosta 2010). This surprising adoption of the concept across theoretical faultlines can be argued to be the result of what is essentially a misunderstanding: GVCs' adoption was enabled by an inherent ambiguity regarding their theoretical underpinnings (Durand, Flacher, and Frigant 2018). This has resulted in an immense diversity of understandings of *what a GVC really is.* We will not attempt to exhaustively describe the different views which have emerged[2], but we will spend the next paragraphs clarifying *what a GVC is* and *what it is not* in the context of this work, which we believe is an essential requirement for clarity of the rest of the paper.

On one hand, there is a segment of the literature for which GVCs provide a general framework to describe: "*the full range of activities, including coordination, that are required to bring a specific product from its conception to its end use and beyond. This includes activities such as design, production, marketing, distribution, support to the final consumer, and governance of the entire process*" (Gibbon and Ponte 2005, p. 77). This kind of conceptualisation views GVCs as a framework which can be used to describe international economic exchanges *lato sensu.* While empirical research performed using such a conceptualisation of GVCs has proved to be a useful tool, apt at describing the specific institutional and political arrangements of international production, it also forgoes describing the specificity of contemporary internationally splintered production. Indeed any kind of interregional economic integration, even as simple as international sourcing of primary commodities fits in that framework.

In contrast to this view, we will inscribe our use of the notion of GVCs in another tradition, one which

[2]For an overview of the litterature, we recommand the reading of Durand, Flacher, and Frigant (2018).

views GVCs as a specific form of industrial organisation (Milberg and Winkler 2013; Carballa Smichowski, Durand, and Knauss 2021), with corresponding implications regarding the structuration of the international division of labour and the distribution of value created in the chain. In this work, we accept that "*a GVC delineates a geographically - and often also legally-fragmented economic space where incomplete commodities are functionally integrated and valorised through a unified labour process*" (Carballa Smichowski, Durand, and Knauss 2021, p.2). Within this space, incomplete commodities are integrated to produce completed commodities. A given chain is thus identified by the final commodity it produces, in other words, in terms of final demand. What is key here is that the incomplete commodities exchanged within the chain are intrinsically complementary. This integration of the different complementary commodities – orchestrated by the *lead firms*[3] which shape the organisation of the production process – has a concrete consequences: the intermediate goods exchanged within the chain cannot be valorised in the same way they are inside the chain outside of the chain. This gives the lead firm the ability to capture a comparatively higher share of value than the other actors in the chain, as they are the only possible buyers for some of the goods which are traded within the boundaries of the chain[4]. The *lead firm* thus holds a comparatively higher level of market power within the space of the GVC than all subcontractors, and the GVC's borders are defined by the limits of exertion of this market power. This asymmetry of market power does not only define the boundaries of the chain, but it also characterises the distribution of value amongst actors of the production network. A fruitful literature has emerged around the observation of the phenomenon of *polarisation* of the *capture* of value-added within the chain (Stöllinger 2021; Shin, Kraemer, and Dedrick 2012; Rungi and Del Prete 2018). The stylised fact which emerges is that the activities which capture most of the added value are the ones which either happen extremely early in the production process (R&D, design) or very late (retail, marketing) at the expense to the ones happening in between (production, logistics). This is often referred to as a "*similing curve*" of *value capture*. Defining GVCs as a specific form of industrial organisation has the advantages of allowing us to clearly delineate borders to GVCs, ones that go beyond the legal formalism of intra-firm trade and arm's-length relations. This is key for constructing indicators able to distinguish GVC trade, ones which avoids conflating the general notion of an increased preeminence of trade with the specific

[3]The quintessential lead firm example being Apple, acting as the coordinator of production networks such as the one of the iPod, which we discussed *supra*.

[4]Think, for example, of our iPod example. Even if the iPod casing is manufactured by a subcontractor of Apple, and there is therefore no opportunity for transfer pricing, Apple is the sole possible buyer for that good and holds an oligopsony which should be expected to drive the price down.

modern, splintered character of international production (Carballa Smichowski, Durand, and Knauss 2021), which is what we attempt in section 5.

3 Input-Output Approach, Methodology and Data

In the following section, we adopt the following notation: matrices are written out in capital Greek or Latin letters (for instance M), vectors (row or column) are written in lower case bold letters (for instance $\boldsymbol{v}$), and scalars are written in lowercase, regular weight letters (for instance s). We index matrices as follows: let M_{ij} denote the element of the matrix M positioned in the row i and the column j, and let $M_{\cdot j}$ and $M_{i \cdot}$ respectively denote the row-vector constructed by isolating the j-th row of M and the column-vector constructed by isolating the i-th column of M. For vectors, we let v_i be the i-th element of the row (or column) vector $\boldsymbol{v}$. Given some column or row vector $\boldsymbol{v}$, $\hat{\boldsymbol{v}}$ denotes the square matrix with diagonal elements $\hat{v}_{ii}$ corresponding to the elements $\square_i$ of vector $\square$ and 0 everywhere else.

Our analysis considers a multi-sectoral, multi-regional[5] economic model which satisfies three key assumptions:

A.1) Stationary economy: total production of every sector of the economy is exactly absorbed by the combination of final demand and intermediate demand of other sectors.

A.2) Single commodity per sector: we assume that the goods produced by each sector are homogeneous. This means that buying p of output of, say, the Chinese machine-tool production sector is always associated with the same productive activity regardless of the identity of the buyer or the quantity bought.

A.3) Linear technological coefficients: we assume that the production of a unit of output of some industrial sector always require the same share of intermediate good inputs and labour from various other sectors. Inputs are not substituted at the sectoral level.

[5] While our setup is such that we may have a sector repeated in two different regions (for instance, we have a Spanish and a German automotive sector), from here on, for conciseness, we will simply refer to “a sector” to refer to “a sector in a region”. This means that a delivery from the German automotive sector to the Spanish automotive sector is referred to as an inter-sectoral delivery (even though it is both inter-sectoral and inter-regional).

The Open Leontief system Let Z be a $(n \times n)$ square matrix with elements Z_{ij} measuring the inter-sectoral intermediate deliveries (measured in monetary units) from industry i to industry j. The $(n \times 1)$ column vector $\boldsymbol{f}$ with element f_i representing final demand (measured in monetary units) of the production of sector i. The $(n \times 1)$ column vector $\boldsymbol{x}$ with element x_i representing total output (measured in monetary units) of sector i. The $(1 \times n)$ row vector $\boldsymbol{l}$ with element l_j representing the labour (measured in number of equivalent full-time workers, labour hours, or some other temporal unit) expanded in the sectors j. Finally let us introduce the $(n \times 1)$ column summation vector $\mathbf{1}_{n\times 1}$. Because for each sector total output x_i must equal the sum of intermediate demand and final demand $\left(f_i + \sum_{j=1}^{n} Z_{ij}\right)$, this is assumption A.1. Thus, it holds that:

$$Z\mathbf{1}_{n\times 1} + \boldsymbol{f} = \boldsymbol{x}. \tag{1}$$

It is also true that $\boldsymbol{l}\mathbf{1}_{n\times 1} = L_{\text{tot}}$ where $L_{\text{tot}} \in \mathbb{R}^{+}$ is the total time worked in our economy. Let us further introduce the $(n \times n)$ technological coefficient matrix A with elements $A_{ij} = Z_{ij}/x_j$ and the $(1 \times n)$ labour coefficient row-vector $\boldsymbol{a}$ with elements $a_i = l_i/x_i$. We call the tuple $(A, \boldsymbol{a})$ the *technology* of our economy, as it containts all information regarding to intermediate goods and labour usage required for production in every sector. This interpretation is meaningful because of assumption A.3. It is easily verified from (1) and the definition of A that the following equation holds:

$$A\boldsymbol{x} + \boldsymbol{f} = \boldsymbol{x}. \tag{2}$$

Similarity it is easily verifiable from the definition of $\boldsymbol{a}$ that $\boldsymbol{a}\boldsymbol{x} = L_{\text{tot}}$. The equation (2) can be rearranged as $(I - A)\boldsymbol{f} = \boldsymbol{x}$ which, assuming that $(I - A)$ is full rank[6] yields a unique solution for total production, given final demand:

$$\boldsymbol{x} = (I - A)^{-1}\boldsymbol{f}. \tag{3}$$

We call the term $(I - A)^{-1}$ the *Leontief inverse*, let us denote elements of this matrix as α_{ij}. The economic interpretation of the *Leontief inverse* is the following. The element α_{ij} represent the share (in

[6] Which is an almost certain property for a technological coefficient matrix, to see why refer to the Chapter 3 of Pasinetti (1977).

relative monetary units) of output of the i-th industrial sector required for the production of one unit of final demand of the j-th good, taking into account all the interconnections within the economy.

Decomposing vertical integration Turning our attention back to the labour embodied in production. Let us discuss how to make use of the Leontief inverse to understand how much total labour is *embodied* in the final-demand output a given industry. Under assumptions A.1,2,3, this quantity, which we refer to as the vector of *vertically integrated labour* is given by the $(1 \times n)$ row-vector $\boldsymbol{\lambda}$, which is computed as:

$$\boldsymbol{\lambda} = \boldsymbol{a}(I - A)^{-1}\hat{\boldsymbol{f}}, \tag{4}$$

Each element λ_i of the vector $\boldsymbol{\lambda}$ gives the total amount of labour required to produce the final demand for sector j (see appendix A for a more detailed derivation). This is an interesting quantity, in the sense that it gives us a way to measure the total labour hours invested in each sector's output (accounting for interconnection between industries across geographical boundaries) but it is not sufficient for us to answer our research question. For this we need to decompose into which sector the labour embodied in final demand production comes. This is given by the $(n \times n)$ matrix Λ, which is computed as:

$$\Lambda = \hat{\boldsymbol{a}}(I - A)^{-1}\hat{\boldsymbol{f}}, \tag{5}$$

The elements of the matrix Λ have the following economic interpretation. The element Λ_{ij} gives the quantity of labour expanded in sector i for production of the final demand of sector j (see appendix A for a more detailled derivation). The matrix Λ gives us the perfect tool to approach our research question. Indeed, the vector $\Lambda_{\cdot j}$ (the j-th column of Λ) gives us the quantity of labour expanded in each sector for the production of final demand output of sector j.

Decomposing value-added contributions Let us now introduce the $(1\times n)$ row vector $\boldsymbol{v}$ of the value-added in each sector, and the corresponding value-added coefficient vector $\boldsymbol{\nu}$, with elements $\nu_i = v_i/x_i$. The same method can then be used to decompose the matrix of value-added contributions of each sector, which is given by:

$$V = \hat{\boldsymbol{\nu}}(I - A)^{-1}\hat{\boldsymbol{f}}, \tag{6}$$

where the element V_{ij} of the matrix V giving the value-added yielded in sector i by the production of final demand for sector j.

Data We make use of FIGARO-E3 (Cazcarro et al. 2025), a high-resolution extended multi-regional input-output (MRIO) database constructed with the aim of providing MRIO data consistent with national accounts. The last (and so far the unique) version of the database covers the year 2015. FIGARO-E3 is particularly relevant for labour accounting because it is one of the few MRIO options to provide highly detailed labour satellite accounts. It is one of the most detailed MRIO databases available for the European area, including all EU member-states as well as non-EU member European countries (Switzerland, Russia, etc.) and noteworthy non-European economies (China, the United States, Japan, etc.). In total the database includes 46 regions (45 countries and a *Rest of the World* region containing all remaining countries) and 176 sectors (there is also a product decomposition into 213 products, which we do not make use of in this work). The labour accounts of the FIGARO-E3 database are constructed from the OECD's Trade in Employment database (Organisation for Economic Co-operation and Development 2023) employment is measured in number of equivalent full-time positions (not labour hours) and the data is disaggregated by gender (male and female) and skill level (high, medium and low skill, with skill either categorized by education or by occupation). We do not make use of this skill/gender disaggregation in this work.

<table>
<tr><td colspan="3" rowspan="3"></td><td colspan="7">Use</td><td colspan="3">Final demand</td></tr>
<tr><td colspan="3">Argentina</td><td>...</td><td colspan="3">Rest of the World</td><td rowspan="2">Argentina</td><td rowspan="2">...
...</td><td rowspan="2">Rest of the World</td></tr>
<tr><td>Sector 1</td><td>...</td><td>Sector 76</td><td>...</td><td>Sector 1</td><td>...</td><td>Sector 76</td></tr>
<tr><td rowspan="7">Supply</td><td rowspan="3">Argentina</td><td>Sector 1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>...</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
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<tr><td rowspan="3">Rest of the World</td><td>Sector 1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>...</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>Sector 76</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td rowspan="2">Value-added</td><td colspan="2">Labour</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="2">Capital</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="3">Labour accounts</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>

Table 1: Illustration of the Figaro-E3 MRIO table (with labour accounts) used in the context of this work.

We perform a series of preprocessing steps on the database and its associated satellite accounts to prepare the data for our embodied labour analysis. The first is to aggregate the sectors from the 176 disaggregated sectors of the FIGARO-E3 disaggregation to the 76 sector disaggregation of the NACE[7] classification (Eurostat 2008). We perform this step for three reasons, first, the extreme level of disaggregation of the dataset is mostly relevant for performing computation with environmental satellite accounts[8] which we do not perform here. Second, as is indicated in Cazcarro et al. (2025) the disaggregated labour accounts may not be very reliable as they are obtained by desegregating the much less granular labour accounts from the OECD. Third, reducing the number of sectors reduces numerical error in the matrix inversions which we need to perform. Our second pre-processing step consists in identifying *empty sectors*: those are sectors that do not exist in certain economies. For instance most countries do not produce oil, which implies that their oil sectors is associated with a line and a column completely filled with 0 values. These empty sectors are also associated with 0 value-added and 0 final demand. The presence of these 0 rows-columns in the input-output database creates difficulties in computing the *leontief inverse* and they are, by definition, disconnected from all the other sectors. So we simply prune them out of the database. At this point there remains a short list of sectors which are problematic. These are sectors with either fully 0 rows or fully 0 columns (i.e. sectors which are disconnected from the other sectors) but that have non-0 final demand/value-added. These cannot simply be removed as it would introduce errors. What we do is that we aggregate them to the "functionally nearest" sector in the NACE classification (we aggregate manufacturing sectors with other manufacturing sectors, primary sectors with other primary sectors, etc.). Together the aggregation of the sectors and the removal of *empy sectors* bring the total number of sectors from 8096 down to 3363. We detail the exact list of sectors which we prune and aggregated in the appendix C. The structure of our pre-processed table is represented in table 1.

Once our data is pre-processed we compute tables decomposing the origin of *embodied labour* and value-added using this dataset using equations (5) and (6). These tables are (3363×3363) square matrices with columns representing respectively the embodied labour contribution of each sector to each other sector's final demand production, and the value added extracted by each sector for each other sector's final demand production. As per our theoretical framework, GVCs are identified by the finished commodity they produce. As each column of the tables are associated with the final demand of a given sector, each

[7]Which is the classification used by the yearly rolled-out version of the Figaro dataset.

[8]For instance, the large level of disaggregation in agriculture allows to properly distinguish greenhouse gas intensive sectors such as animal husbandry from less intensive ones such as rice farming

<table>
<tr><td colspan="3" rowspan="3"></td><th colspan="7">Labour/VA Use</th></tr>
<tr><td colspan="3">Argentina</td><td>...</td><td colspan="3">Rest of the World</td></tr>
<tr><td>Sector 1</td><td>...</td><td>Sector 76</td><td>...</td><td>Sector 1</td><td>...</td><td>Sector 76</td></tr>
<tr><th rowspan="7">Labour/VA Supply</th><td rowspan="3">Argentina</td><td>Sector 1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
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<tr><td>...</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td rowspan="3">Rest of the World</td><td>Sector 1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>...</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>Sector 76</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>

Table 2: Illustration of the labour/value-added decomposition tables by our method. The rows correspond to suppliers and the columns to usage. For instance the top left element of the table in its labour version measures the quantity of labour expanded by the first sector of the Argentinian economy for the production of the final demand of goods from the first sector of the Argentinian economy.

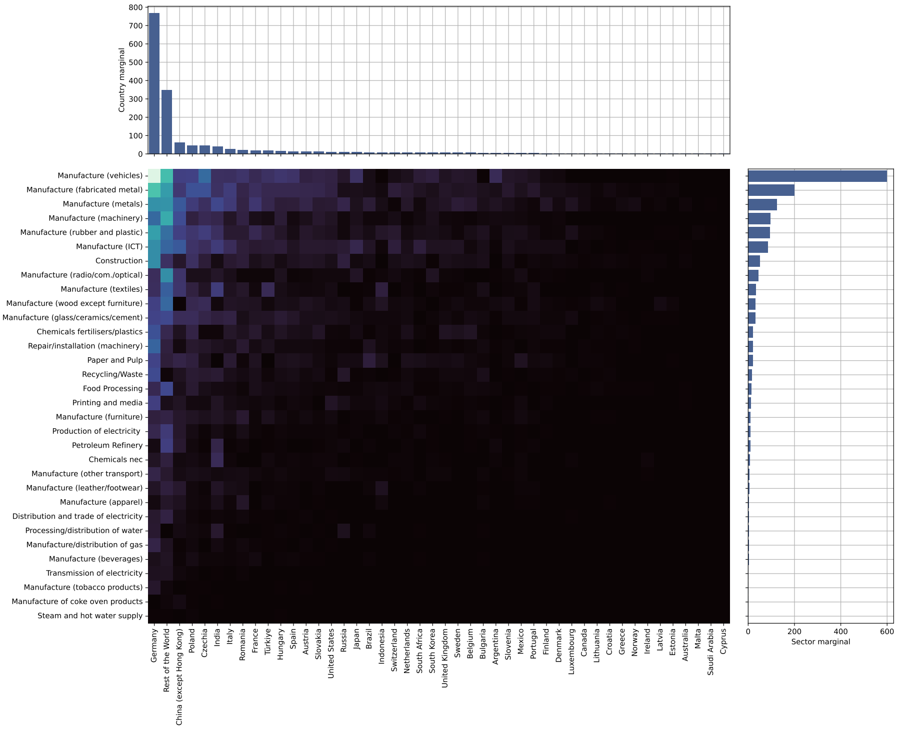
column of a given decomposition table respectively gives the value-added or *embodied labour* decomposition corresponding to a specific GVC. The tables, as well as our replication code, are publicly available at the following address: https://github.com/RenardDesNeiges/embodied_labour_figaro_e3.git.

4 A Case Study: the German Automotive Sector Industry

To assess the utility of our *embodied labour* measurements for understanding GVCs we apply our framework to a case study of the German automotive sector, this specific case study is a classic example for showcasing the utility of input-output analysis (Timmer et al. 2015). It allows us both to test claims made by Sturgeon, Van Biesebroeck, and Gereffi (2008), Timmer et al. (2015), and De Ville (2018) regarding the pattern of global value chain integration of the (German) automotive sector and to evaluate whether labour-based measurements reveal structural features that value-added indicators cannot.

Our methodology consists in considering both the *embodied labour* and the value-added decompositions of the German automotive GVC. This value chain is identified by the German C29 sector (in the NACE classification) and is studied by extracting a single column of of the corresponding decomposition tables (see Table 2). In the context of this analysis, we focus on the case study of Germany, but also use data from Spain and the United Kingdom as points of comparison. Following the principles introduced in section 2, and in order to perform an analysis consistent with GVCs (and not with trade *lato sensu*), we exclude primary commodities[9] from our analysis (see section 2), as we are certain that they are not functionally integrated in the chain. Furthermore, because our case study focuses on production steps

[9]That is, primary sector output.



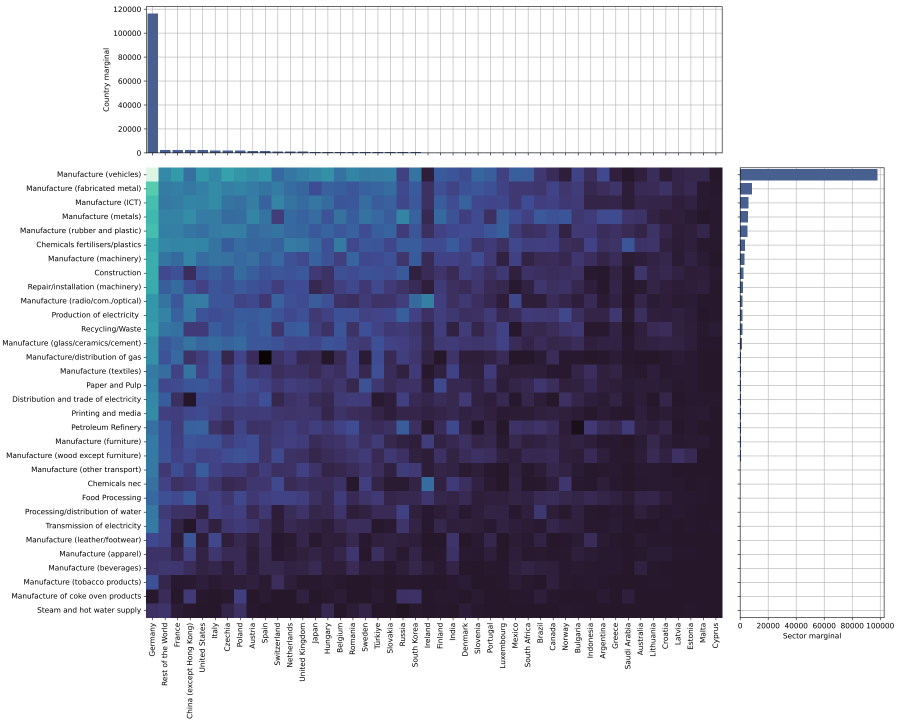


Figure 1: **Top:** decomposition of embodied labour in the output of the German automotive value chain (‘*Manufacture of motor vehicles, trailers and semi-trailers*’, sector C29 in the NACE classification). Each pixel of the heatmap represents a given sector (the row) in a given region (the column); the colour map is log-normalised for increased readability of the small values. This is a visualisation of the column $\Lambda_{\cdot,j}$ of the labour decomposition matrix. The two bar graphs give the “*marginals*”, horizontally: the embodied labour by country, regardless of the sector and vertically: the embodied labour by sector, regardless of the country. All primary and tertiary sectors are omitted, and the sectors and countries are sorted by order of importance. **Bottom:** the exact same decomposition for value-added per sector/region. This time this a column of the matrix $V_{\cdot,j}$.

we exclude distribution and services, this has the interest of removing a large part of activity happening in offshore financial centers from our data, and, we believe gives a more legible picture of the German automotive GVC.

According to De Ville (2018), Germany's automotive industry is characterised by *nearshoring*. This refers to a practice where firms choose to offshore some of their production to geographically proximate countries rather than to distant low-cost regions. This is understood to allow firms to: reduce coordination costs associated with distance, preserve comparative institutional advantages rooted in non-market coordination and thus maintain quality-differentiated production while still benefiting from lower production costs in nearby regions. In the case of Germany, the nearshoring destinations identified by De Ville are located in Eastern and Central Europe. Sturgeon, Van Biesebroeck, and Gereffi (2008) on the other hand note that the automotive sector in general is characterized by a trend of *regionalization* whereby regional clusters specialize in different areas domain of production. We will consider both of these claims through the lens of our *embodied labour*/value-added decomposition methodology.

The decomposition of the German automotive GVC is presented in Figure 1. The figure shows, on top, the total amount of labour embodied in the final demand output of the German automotive sector across countries and sectors (excluding the primary and service sectors), and below a visualisation of the same decomposition performed in value-added. The first, very obvious stylised fact which emerges from this data is that value-added is much more concentrated than labour, both with respect to a sectoral or a regional decomposition. This is coherent with the notion that there is asymmetric capture of value-added within the automotive production value chain. This asymmetry is clearly not only playing in favour of German companies, indeed the ordering of the largest contributor-regions to the sector changes whether measurement is performed in value-added or in *embodied labour*. The 5 largest *embodied labour* contributors to the German automotive sector being Germany itself, the Rest of the World aggregate region, China, Poland and the Czech Republic while the largest contributors in value-added are: Germany itself, the Rest of the World aggregate region, France, China and the United States. It is striking that measuring contribution in value-added instead of labour removes the Eastern European countries from the top 5 (Poland and the Czech Republic) in favour of France and the United States. This provides a first demonstration that *embodied labour* provides a fundamentally different picture of where production actually occurs.

This result directly addresses the motivation of this work, that is, that labour reveals the production structure that value-added hides. Because value-added is heavily shaped by mark-ups, transfer pricing and the location of high-profit intangible activities, it artificially inflates the domestic importance of economies of countries in which lead-firms are based, such as Germany, in our example, but also other advanced economies such as Germany or the United States. In contrast, labour is not subject to such price distortions: the full employment count provided by Figaro-E3 provides a proxy for how much time has been allocated for production in a given sector in a given region, regardless of the associated value-added figures.

	Germany		**UK**		**Spain**	
	Labour	VA	Labour	VA	Labour	VA
Domestic	49.19%	80.07%	37.21%	82.85%	22.51%	49.01%
Eastern Europe	9.76%	4.19%	1.70%	1.92%	6.66%	5.16%
Northern/Western Europe	4.27%	6.20%	3.94%	17.15%	10.95%	25.91%
Southern Europe	2.77%	2.42%	0.93%	2.30%	3.76%	4.67%
NAFTA	1.00%	1.64%	4.86%	25.33%	0.95%	2.25%
China (except Hong Kong)	3.98%	1.57%	12.03%	15.86%	5.88%	3.84%
India	2.58%	0.18%	9.36%	1.98%	3.93%	0.46%
Rest of the World	26.45%	3.73%	67.17%	35.47%	45.37%	8.71%
Total	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%

Table 3: Regionally aggregated labour and value added contribution for the German, Spanish and British automotive sectors (C29 in the Figaro classification). The contributions of the different aggregate regions in labour and value added are computed from the columns $\Lambda_{\cdot,j}$ and $V\cdot, j$ obtained from the labour and value added decomposition. The regions are aggregated as follows. Southern Europe: Italy, Spain, Portugal, Greece, Malta, Cyprus, Spain; NAFTA: United States, Mexico, Canada; Northern/Western Europe: France, Austria, Switzerland, Netherlands, United Kingdom, Sweden, Belgium, Finland, Denmark, Luxembourg, Norway, Ireland, Germany; Eastern Europe: Poland, Czechia, Romania, Hungary, Slovakia, Bulgaria, Slovenia, Lithuania, Croatia, Latvia, Estonia.

	Germany		**UK**		**Spain**	
	Labour	VA	Labour	VA	Labour	VA
Eastern Europe	19.22%	21.01%	1.70%	1.92%	8.59%	10.11%
Northern/Western Europe	8.40%	31.12%	3.94%	17.15%	14.13%	50.82%
Southern Europe	5.45%	12.15%	0.93%	2.30%	4.86%	9.15%
NAFTA	1.97%	8.22%	4.86%	25.33%	1.22%	4.42%
China (except Hong Kong)	7.83%	7.90%	12.03%	15.86%	7.59%	7.53%
India	5.08%	0.89%	9.36%	1.98%	5.07%	0.89%
Rest of the World	52.06%	18.71%	67.17%	35.47%	58.55%	17.08%
Europe	33.07%	64.28%	6.57%	21.37%	27.58%	70.08%
Total	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%

Table 4: Aggregate contributions, of different regions to the automotive sectors of Germany, the United Kingdom (UK) and Spain, in terms of labour and value added (VA) embodied in intermediate product imports. The percentage share of contribution excludes the domestic labour and value-added contributions to the automotive sectors in question. The Europe row is the sum of the three European subdivisions.

Table 3 decomposes the labour and value added contribution of different regions to German automotive production. It shows that the automotive supply chain relies far more on foreign productive activities than what would be suggested by value-added statistics. Indeed, our computations show that Germany accounts for only 49.19% share of the labour required to produce its automotive output, but nonetheless reaps 80.07% of the added value. In the aggregate, the foreign sector is contributing slightly more labour than Germany itself. A clearer picture of how labour-based indicators modify the interpretation of Germanys position in global value chains emerges when we relate our regional labour distribution to the value-added patterns documented by De Ville (2018). De Ville shows[10] that Germanys foreign value-added sourcing is, comparatively with other automotive producers, such as the UK or the US[11], more regional, with the EU supplying the clear majority of foreign intermediates commodities used by the sector. Table 4 confirms that in labour terms Germany is relatively more regionally oriented than the UK, but this relative regional orientation must be nuanced: most of the labour embodied in German car production is actually imported from outside Europe. With 33.07% of the foreign labour embodied in German automotive production originating within the EU, the degree of regional integration of the German automotive sector is far exceeding that of the United Kingdom's (5.57%), and lightly exceeding that of Spain's (27.58%), but it is nonetheless an absolute minority of the imported *embodied labour* of the sector.

The labour-embodied approach substantially extends and nuances the kind of findings that can be obtained when concentrating exclusively on value-added, such as what Timmer et al. (2015) and De Ville (2018) perform. While a measurement by value-added would suggest that Europe is responsible for 64.28% of the imports of intermediate goods by the German automotive sector, suggesting that the German automotive GVC is majority-European, measurement in embodied labour, shows that this is not the case. Table 4 shows that Rest of World is the single largest source of imported embodied labour. China appears as the third-largest labour source, ahead of any individual European partner except Germany itself. While value-added based analysis identifies Asia as a second-rank supplier, our labour decomposition shows that China plays a central role in supplying goods used for producing cars eg. electronics, metals, chemicals and rubber (see Figure 2). This indicates that the functional division

[10]See De Ville (2018) Table 2 and 3.

[11]While DeVille's case study involves a comparaison with the US we choose to concentrate only on European countries as the coverage of the trade partners of the United States is not as good as that of European countries in our dataset.

of labour in German automotive production is more geographically dispersed than suggested by value-added indicators alone. Labour-embodied methods therefore highlights Asian sectors that are obscured when using value-added alone (some of which are aggregated in the *Rest of the World* region because of the European-centric nature of our dataset).

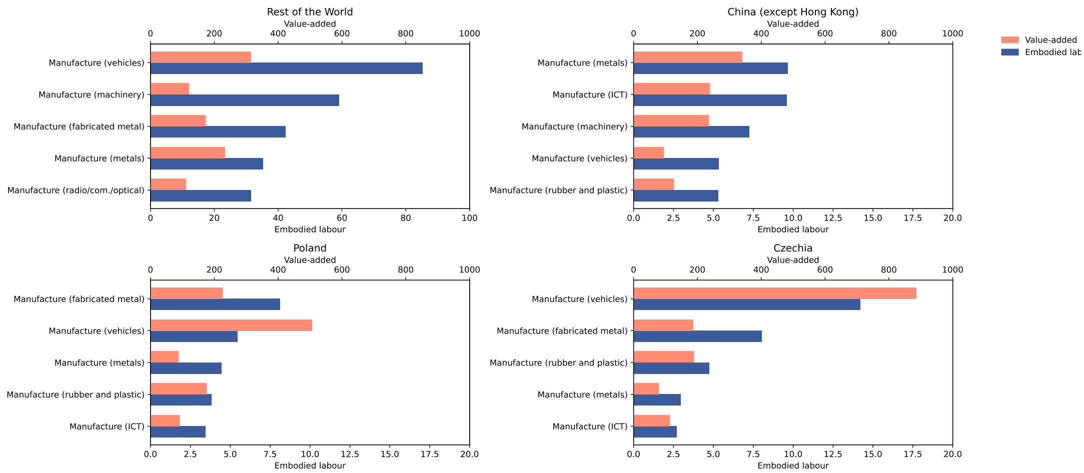


Figure 2: Top foreign contributors of embodied labour to the German automotive sector's final demand production. Here we show the top 5 contributing sectors of each of the top 4 foreign contributing regions (by "top contributing", we mean that contribute most *embodied labour*). Value added is measured in mio and labour is measured in thousand of equivalent full-time employed workers. All x axes are scaled identically except for foreign labour which has significantly larger values than the other regions.

Further disaggregating at the European level, we see that the largest European labour contributor to German imports is Eastern Europe (Table 4), but that this is not the case when measuring in value-added. In the value added measurement, Nothern/Western Europe appears more significant. Building on this regional pattern at the aggregate level, the more detailed country-level decomposition in Figure 1 aligns with De Villes account of how this regional integration is structured internally. His value-added analysis emphasises that German automotive firms source a large share of their intermediates from Central and Eastern Europe, and our labour-based results reinforce this claim. Poland, the Czech Republic and related economies contribute substantial embodied labour shares, partially validating the *nearshoring* hypothesis of De Ville and illustrating how these economies have taken up some labour-intensive[12] stages of production that complement Germanys high-skill, capital-intensive domestic activities have

[12]Our results only show that a lot of labour is expanded for the German automotive GVC in these countries, but have not investigated whether the labour intensive activities performed are also capital intensive (as this is not a question which our dataset can answer by itself). While neoclassical trade theory (specifically the Heckscher-Ohlin-Samuelson model) would suggest that they must be less capital-intensive, they are actually activities that are classically understood to be quite capital-intensive as well (vehicle manufacturing, machine tool manufacturing, metallurgy). As Milberg and Winkler (2013) and Kaczmarczyk (2023) note, modern internationally splintered production with high capital mobility and persistent current account imbalances has put into question the validity of these approaches. This question would warrant further research.

abandonned.

A more detailed analysis of the sectoral decomposition of the *embodied labour* imported from these countries shows diverse specialisation patterns (Figure 2). The contribution of the *Rest of the World* aggregate sector is the largest by far and is characterised by high labour content relative to added value. Chinese imports are led by manufacturing of metals, by the ICT sector and by machinery production. This is in contrast with the imports from Czechia and Poland in which the ICT sector does not play such a large role, but the vehicle manufacturing sector does, suggesting that these countries are involved in the more downstream stages of the GVC. This regionalisation pattern tracks with the clustering dynamics identified by Sturgeon, Van Biesebroeck, and Gereffi (2008) and, to some extent, corroborates the regional pattern identified by De Ville (2018), but also shows how it manifests across individual European partner countries.

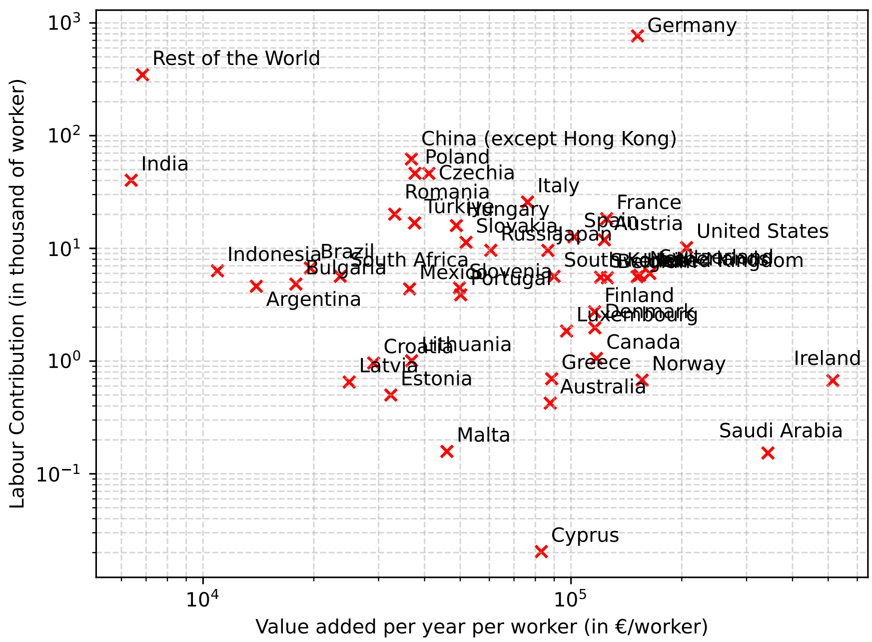


Figure 3: Value added per year per worker (in /person) vs number of workers (in thousands of workers) attributable to the final demand of the German automotive sector per region. When excluding the domestic contributions, we find a correlation of -0.20 between the labour contribution (in thousands of workers) and the value added per year per worker.

Our findings reveal that while the German automotive value chain is comparatively more regionally concentrated ("*nearshored*") than other European automotive value chains, it is also very much globally labour-dependent. Across our dataset, there is a clear negative correlation between labour contribution and low value-added per worker (see Figure 3), which is consistent with the idea of a race to the bottom in terms of input costs (Ali 2015; Nathan and Sarkar 2011) and has the effect of obfuscating the geography of production. Indeed, the global dimension of the German automotive GVC is largely invisible in the value-added measures, which understate contributions from low-wage economies because these economies fail to capture much of the total value-added generated in the value chain. This is especially clear when considering the low value-added to embodied labour ratio of the top labour contributing sectors of the *Rest of the World* region shown in Figure 2. This, of course, raises more general questions about patterns of integration of labour into GVCs. While our dataset is sufficiently disaggregated to be used for econometric studies of the labour/value-added relations along GVCs we have not gone down this investigation road and leave this question for further studies.

In conclusion, our case study demonstrates that labour-embodied indicators provide a more realistic and structurally transparent account of German automotive global value chains than value-added measures. It confirms the nearshoring pattern identified by De Ville (2018) and the clustering phenomenon discussed by Sturgeon, Van Biesebroeck, and Gereffi (2008) while simultaneously revealing a deeper global labour dependence and showcasing the strong polarisation of the capture of added value. This is consistent with the findings of the *smiling curve* literature (Rungi and Del Prete 2018). Embodied labour measurements thus not only complement but also significantly revise the conclusions drawn from the sole use of added value measurements.

5 Embodied Labour Indicators of GVC Participation and Value Capture

We argue that there are two key aspects of GVC integration worth measuring with country-level indicators. First GVC participation, i.e. to what extent a country is involved in GVC trade, second GVC benefits, i.e to what extent a country benefits from its participation in GVC trade. In the following subsection, we propose labour-centric indicators for both of these aspects, which complement the corresponding added value indicators proposed by Carballa Smichowski, Durand, and Knauss (2021).

The standard measurement of GVC participation in the literature stems from the so-called *two borders rule*: GVC trade is the portion of trade which covers at least two borders (Hummels, Ishii, and Yi

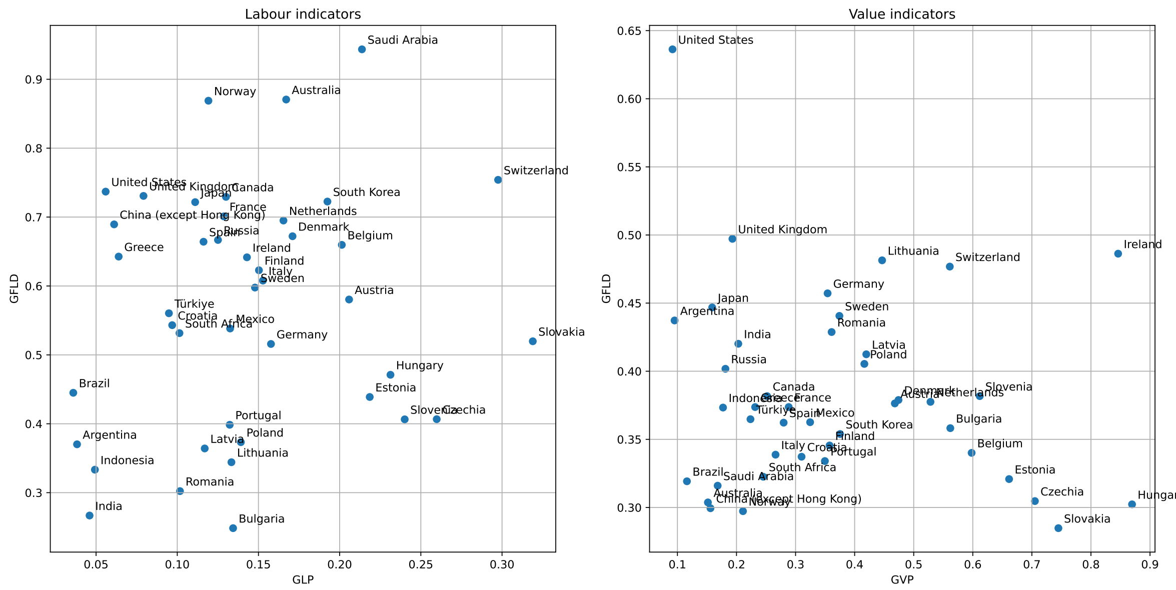


Figure 4: Scatter plot positioning the countries of our dataset according to the indicators discussed in the section. On the left: *embodied labour* indicators, the x-axis measures participation and the y-axis measures foreign labour dependency. On the right: *embodied labour* indicators, the x-axis measures participation and the y-axis measures *value capture*.

2001). Participation is measured through a *backwards* component and a *forwards* component. The backwards component (VS) measures the foreign value-added content of total exports, and the forward component (VS1) measures how third countries rely on the country's export. Carballa Smichowski, Durand, and Knauss (2021) argue that, on the one hand, this approach fails at distinguishing GVC and non-GVC trade and on the other that the use of the *two borders rule* actually excludes a trade which may be part of a GVC. Indeed, from the point of view of the conceptualisation of GVCs as a form of industrial organisation, the length of the chain is not its defining component; instead, what delineates the GVC is the functional integration within the chain. There is therefore no theoretical motivation for excluding trade, which happens across less than two borders. On the other hand, there is trade which we know cannot be part of a GVC: primary sector outputs. Indeed, primary commodities are by definition not functionally integrated, and they are easy to separate from the rest of trade as primary sector is well identified in industrial classification systems (for instance, in the NACE system, the primary sector is identified by industry-codes begining with the letter A or B). Building on these insights Carballa Smichowski, Durand, and Knauss (2021) propose two indicators for measuring *GVC* *participation* (GVP) and *GVC value capture* (GVC) which is the GVC benefit indicator in their system:

$$\mathrm{GVP} = \frac{\mathrm{XDVA_{np}} + \mathrm{iPVM_{np}}}{\mathrm{GDVA}}, \tag{7}$$

$$\mathrm{GVC} = \frac{\mathrm{XDVA_{np}}}{\mathrm{XDVA_{np}} + \mathrm{iPVM_{np}}}. \tag{8}$$

Where $\mathrm{XDVA_{np}}$ denotes the domestic (non-primary sector) value-added in a given country's (or sector's) exports, $\mathrm{iPVM_{np}}$ denotes the gross imports of (non-primary sector) intermediate products (in monetary units) and GVA the country's (or sector) gross value-added. The motivation for the construction of GVP (the participation indicator) is that it gives an upper bound on the relative importance of GVC trade (some of that trade measured by the $\mathrm{XDVA_{np}} + \mathrm{iPVM_{np}}$ component may not be functionally integrated but all functionally integrated commodities traversing the country are included in this value). Regarding the construction of GVC (the benefit indicator) the idea is that it measures value added acquired from exports relative to the upper bound on GVC trade previously introduced. It thus gives a rough approximation of the share of value extracted from GVC trade.

As we have shown in the case study of section 4, value-added measurements minimise the importance of countries taking part of the value chain, but for low returns in terms of value-added. We propose to overcome this limitation by complementing these indicators with two other indicators, measuring participation in terms not of value-added but *embodied labour*. Unlike Carballa Smichowski, Durand, and Knauss (2021), we exclude not only the primary sector, but also the services. This is because when included, services (especially) register as importers of a large amount of foreign embodied labour. While it could be argued that indeed offshore financial centres are big participants in GVCs, we argue that the inclusion of services obfuscates the specificity of GVC participation[13]. Indeed, international financial activities are not a feature specific to the GVC form of industrial organisation. We therefore choose to completely remove services from our computation [14].

[13]Computing our participation indicator, including the service sector, has the effect of replacing the top participating countries by offshore financial centres (Ireland, Belgium, the Netherlands, but not Switzerland). See Figure 7 in appendix B.

[14]Another possible approach, proposed by Basu and Foley (2013), would be to remove the FIRE (Finance, Insurance and Real Estate) sectors.

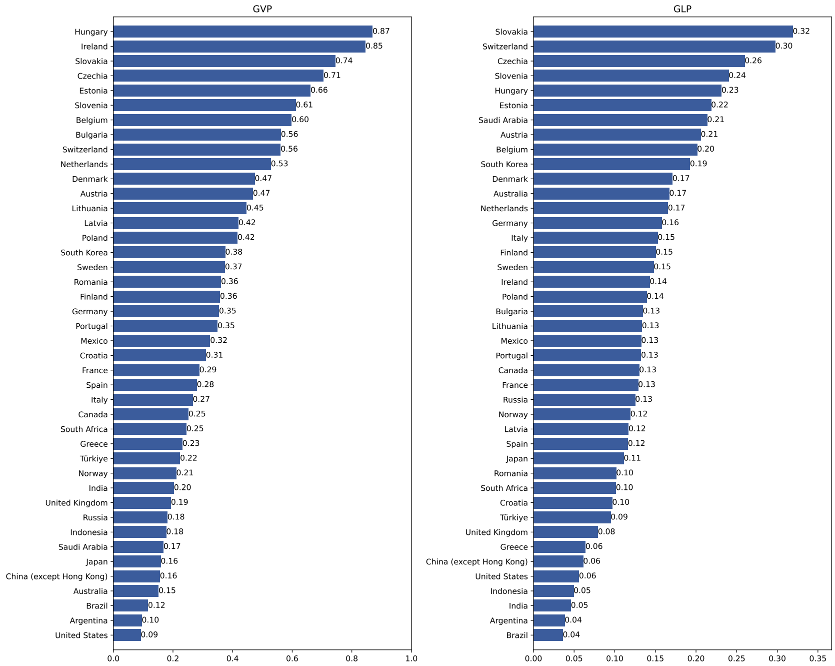


Figure 5: GVC participation indicators computed in value-added according to Carballa Smichowski, Durand, and Knauss (2021) (GPV, left) and in embodied labour according to our proposed methodology (GPL, right). All data is obtained from the processed Figaro-E3 dataset (even for reproducing Carballa Smichowski, Durand, and Knauss (2021)'s value-added results).

Our first indicator indicator measures *GVC participation* (GPL) in non-monetary units:

$$\mathrm{GLP} = \frac{\mathrm{XDL_{nps}} + \mathrm{iPLM_{nps}}}{\mathrm{GDL}}. \tag{9}$$

Where $\mathrm{XDL_{nps}}$ denotes the (non-primary, non-service sector) domestic labour embodied in the country's (or sector's) exports, $\mathrm{iPLM_{nps}}$ denotes the gross imports of (non-primary, non-service sector) intermediate products (in embodied labour units) and GDL the "*gross domestic labour*" (the total amount of labour expanded domestically, regardless of its final demand usage). As is visible in Figure 5, our labour-centric indicator tends to rank higher for small exporter countries deeply tied to large supply chains (Slovakia, Switzerland, Czechia), while the lower-scoring countries are either relatively autarkic countries (China, the United States) or primary sector-intensive countries (Brazil, Argentina). Both indicators are strongly correlated: strongly participating countries in *embodied labour* are strongly participating countries in value-added[15]. One interesting case is the discrepancy between the low value-added and high *embodied labour* score of Saudi-Arabia. Decomposing the indicator's component, it appears that this is the result of the country importing relatively very high quantities of *embodied labour* relative to a very small working

[15] See Figure 8 in appendix B.

population.

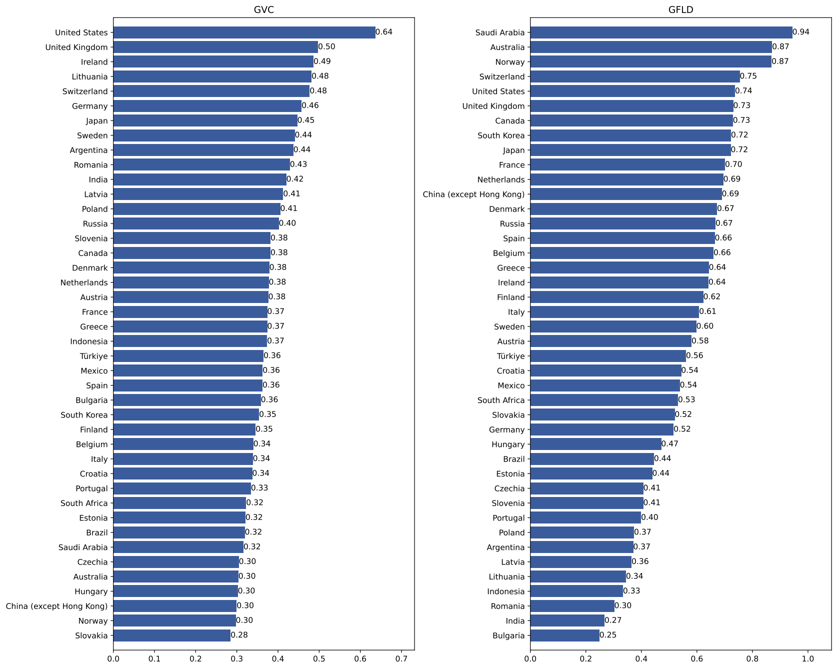


Figure 6: GVC benefit indicators computed in value-added according to Carballa Smichowski, Durand, and Knauss (2021) (GPV, left) and in embodied labour according to our proposed methodology (GFDL, right). All data is obtained from the processed Figaro-E3 dataset (even for reproducing Carballa Smichowski, Durand, and Knauss (2021)'s results).

While our suggested participation indicator essentially mirrors the one proposed by Carballa Smichowski, Durand, and Knauss (2021), simply using embodied labour units instead of value-added, the indicator we propose for GVC benefit is different. We propose to measure *foreign labour dependency* instead of *value capture*. This is given the following expression:

$$\mathrm{GFDL} = \frac{\mathrm{iPLM_{nps}}}{\mathrm{XDL_{nps}} + \mathrm{iPLM_{nps}}}. \tag{10}$$

The idea is that this indicator measures a country's dependence on foreign embodied labour imports relative to its implication in GVC trade. When comparing our foreign labour dependency indicator to the value capture indicator of Carballa Smichowski, Durand, and Knauss (2021), we note that our labour-based proposal yields a slightly more polarised result. Unlike with participation indicators, the correlation between the value-added and the *embodied labour* metrics is less clear[16]. This suggests that dependency on imported foreign labour for GVC production does not automatically come with increased domestic value-added capture. For instance, China is a country which is heavily reliant on foreign labour

[16]See Figure 9 in appendix B.

according to the GFLD indicator, yet scores rather low on the *value capture* index. Conversely, the United States dominates *value capture* but does not score so high on the foreign labour dependency index. One pattern in the data, countries that are exporters of hydrocarbons score abnormally high in terms of foreign labour dependency (Saudi Arabia, for instance). Looking at the component of the indicator, this appears to be because these countries have a very low quantity of domestically expanded labour relative to the volume of foreign embodied labour they import for their production. Countries which are low scorers in the foreign labour dependency index here are not only non-hydrocarbon primary commodity-centred economies (Argentina) but also cheap manufacturing hubs (Indonesia, Bulgaria, Romania, India).

As in the case study from section 4, we do not argue in favour of replacing added value indicators by labour indicators, but rather in favour of using both, as together these indicators are revealing of a level of complexity that could not be captured through value-added alone.

6 Conclusions

This paper proposed and applied a novel methodology for measuring global value chain (GVC) integration using *embodied labour* rather than conventional trade in value-added metrics. By constructing a *embodied labour* decomposition table through input-output analysis, we aimed to address two key shortcomings in existing GVC indicators: their inability to distinguish meaningfully between GVC and non-GVC trade, and their susceptibility to price distortions. Using the FIGARO-E3 multi-regional input-output database, we computed sector- and country-level measures of labour embodiment and compared them with value-added accounts. We showcased the utility of these measurements with two applications: a case study of the German automotive value chain and the construction of labour-based indicators of GVC participation and foreign labour dependency.

Our case study of the German automotive industry yielded several important findings. First, labour-based measurements reveal a significantly more geographically dispersed structure of production than value-added indicators suggest. Following Milberg and Winkler (2013) we argue that this happens because of asymmetries in the capture of value along GVCs, for as a result of oligopolistic/oligopsnistic market structures and transfer pricing practices. In the case of Germanys automotive industry, while value-added metrics attribute over 80% of production to Germany itself, *embodied labour* measurements show that half of the labour required originates abroadprimarily in Eastern Europe, China, and the Rest of the World. This underscores how value-added metrics tend to overemphasise the role of lead-firm economies and understate the contributions of low-wage regions. Second, we showed how our method allows for uncovering specialisation patterns across geographic divisions. Finally, we found a strong negative correlation between labour contribution and value-added per equivalent full-time worker, which is consistent with the *race to the bottom* dynamic in global labour-cost competition.

Next, we introduced new GVC participation (GPL) and foreign labour dependency (GFLD) indicators based on *embodied labour*. These new indicators appear less sensitive to transfer pricing and profit-shifting practices, and thus offer a more structurally transparent view of GVC integration. We argue that their usage, in conjunction with existing value-added indicators, allows for the construction of a more nuanced understanding of integration GVCs.

Despite these contributions, our approach has limitations. The reliance on the FIGARO-E3 database restricts our analysis to the year 2015, rendering impossible any analysis of temporal dynamics. Labour data is measured in employment counts rather than hours, which may obscure variations in work intensity and productivity. Additionally, our input-output framework inherits standard assumptions, such as fixed technological coefficients and linear production functions, which may not fully capture real-world heterogeneity and firm-level strategies.

Future research could extend this work in several directions. Developing a price model to explicitly account for transfer pricing and markup differentials would open the way for testing more detailed hypotheses regarding the relation between value-added, wage and labour along GVCs. It would also allow for the computation of counterfactuals, which may be of interest to policymakers. Incorporating more granular labour datasuch as hours worked, skill levels, and gender disaggregationwould allow for richer analyses of distributional and upgrading dynamics within GVCs. Finally, expanding the temporal coverage to panel data would enable the study of GVC evolution over time, particularly in response to geopolitical shifts, trade policies, and technological change, but would require an important effort of data preparation (as far as we are aware, there is no series of multi-regional input-output models involving labour satellite accounts in open access for researchers this would thus requiring constructing an ad-hoc dataset).

In sum, this paper demonstrates that embodied labour provides a valuable complementary lens to value-added analysis, revealing hidden geographies of production and offering indicators more resilient to price manipulation. We hope that this contribution opens new avenues for research into the international organisation of production.

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A Mathematical appendix

In this short mathematical appendix we go through the derivation of the embodied labour decomposition equations presented in the section 3 of the paper. We first consider the computation of vector of total labour embodied in the final demand output of every industry. Recall that this vector is given by:

$$\boldsymbol{\lambda} = \boldsymbol{a}(I - A)^{-1}\hat{\boldsymbol{f}}.$$

The term $(I - A)^{-1}\hat{\boldsymbol{f}}$ is a square $(n \times n)$ matrix with elements as follows:

$$(I - A)^{-1}\hat{\boldsymbol{f}} = \begin{bmatrix} f_1\alpha_{11} & f_2\alpha_{12} & f_3\alpha_{13} & \dots & f_n\alpha_{1n} \\ f_1\alpha_{21} & f_2\alpha_{22} & & & \\ f_1\alpha_{31} & & \ddots & & \\ \vdots & & & & \\ f_1\alpha_{n1} & & & & f_n\alpha_{nn} \end{bmatrix}.$$

It is thus quite obvious from the form of the matrix $(I - A)^{-1}\hat{\boldsymbol{f}}$ that the value of $\boldsymbol{\lambda}$ is given, for the i-th element of $\boldsymbol{\lambda}$ by:

$$\lambda_i = f_i \sum_j \alpha_{ij} a_j.$$

Recalling that α_{ij} is a coefficient which gives the volume of production in sector i which corresponds to one unit of final demand in sector j (for a proof of this, refer to Pasinetti (1977) chapter 3) and that $a_j = l_j/x_j$ is the labour/output ratio of sector j we easily see that each term $a_j\alpha_{ij}$ gives the labour requirement in sector j corresponding to one unit of final demand in sector i. Multiplying the sum of all the labour requirements (across all industries) by the final demand trivially gives us the labour requirement associated with total final demand in sector i, which is what we were trying to show.

Let us now turn our attention to the disaggregation matrix which is the core of our approach. Recall that this matrix is given by

$$\Lambda = \hat{\boldsymbol{a}}(I - A)^{-1}\hat{\boldsymbol{f}}.$$

This is almost exactly the same form as the one we introduced. The only difference is that instead of left multiplying the $(I - A)^{-1}\hat{\boldsymbol{f}}$ matrix by the row vector $\boldsymbol{a}$, we multiply it by the diagonal matrix $\hat{\boldsymbol{a}}$. What we are left with is thus the $(n \times n)$ square matrix Λ. Its elements are given by:

$$\Lambda_{ij} = f_i \alpha_{ij} a_j. \tag{11}$$

Here the interpretation is even more straight forward than in the total embodied labour case and flows directly from the definition of the three terms On the right-hand side of the equation (which we reminded the reader *supra*).

B Indicator appendix

In the following appendix, we include figures omitted from the main body of text, clarifying the effect of decisions made in the design of the embodied labour indicators.

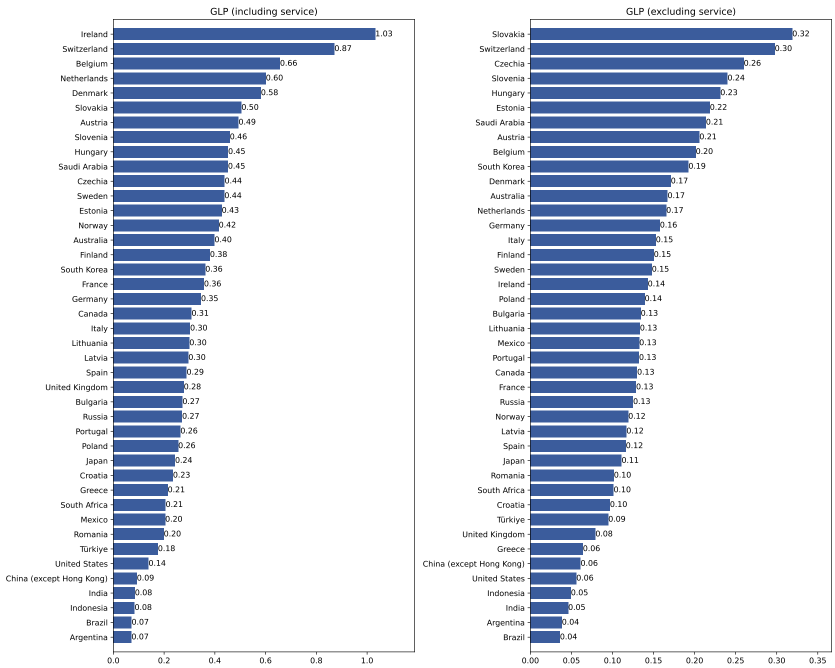

Figure 7: Effect of excluding the service sector in the embodied labour GVC participation (GLP) indicator. It appears that countries known as hubs for the recovery of profits of TNCs (MGI 2012) score abnormally high when the service sector is included.

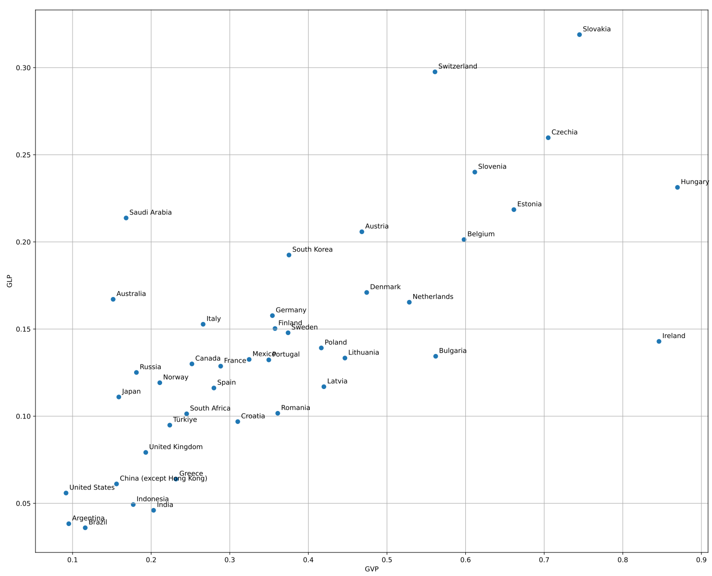

Figure 8: Scatter plot of our *embodied labour* GVC participation indicator v.s. Carballa Smichowski, Durand, and Knauss (2021)'s value-added GVC participation indicator. Note that the indicators are strongly correlated, at the exception of a few countries.

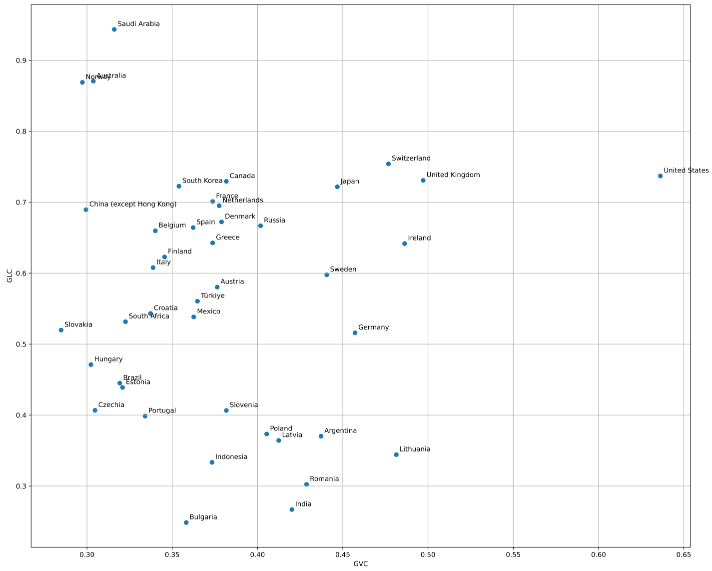

Figure 9: Scatter plot of our *embodied labour* GVC benefit indicator v.s. Carballa Smichowski, Durand, and Knauss (2021)'s value-added GVC benefit indicator. Note that the indicators are less correlated than the participation indicators.

C Data-processing appendix

In this appendix we list the regions and industries included in our dataset.

The list of regions is the following (in alphabetical order): Argentina, Austria, Australia, Belgium, Bulgaria, Brazil, Canada, Switzerland, China (except Hong Kong), Cyprus, Czechia, Germany, Denmark, Estonia, Spain, Finland, France, United Kingdom, Greece, Croatia, Hungary, Indonesia, Ireland, India, Italy, Japan, South Korea, Lithuania, Luxembourg, Latvia, Malta, Mexico, Netherlands, Norway, Poland, Portugal, Romania, Russia, Saudi Arabia, Sweden, Slovenia, Slovakia, Türkiye, United States, South Africa, Rest of the World.

The list of sectors is the standard Figaro industry codes, ordered from primary to services. They are not reproduced here for conciseness (there are 76 industries, some with names taking up multiple lines). But are easily found at the following link: https://joint-research-centre.ec.europa.eu/projects-and-activities/trade-and-industrial-policy-analysis/input-output-accounts/figaro-tables_en.

We list the "*empty sectors*" (sectors with no input, no output, no demand and no added value) which are pruned out of the dataset (Table 5). Finally we list the "*problematic sectors*" (sectors with no input, no output, but some demand and some added value) which are agglomerated with other sectors as well as the sectors with which they are agglomerated (Table 6).

Table 5: List of “*empty sectors*” pruned out of the dataset.

	Country	**Industry**
0	Bulgaria	Private households with employed persons
1	Sweden	Extra-territorial organizations and bodies
2	Slovenia	Manufacture of tobacco products
3	Luxembourg	Tanning and dressing of leather, manufacture of luggage, handbags, saddlery, harness and footwear
4	Indonesia	Extra-territorial organizations and bodies
5	United Kingdom	Extra-territorial organizations and bodies
6	Cyprus	Mining of coal and lignite, extraction of peat
7	Rest of the World	Extra-territorial organizations and bodies
8	Russia	Extra-territorial organizations and bodies
9	Mexico	Extra-territorial organizations and bodies
10	Slovenia	Extra-territorial organizations and bodies
11	Luxembourg	Mining of coal and lignite, extraction of peat
12	Slovakia	Extra-territorial organizations and bodies
13	Malta	Forestry, logging and related service activities
14	South Korea	Extra-territorial organizations and bodies
15	Malta	Extra-territorial organizations and bodies
16	Luxembourg	Fishing, operating of fish hatcheries and fish...
17	Indonesia	Steam and hot water supply
18	Greece	Extra-territorial organizations and bodies
19	Japan	Extra-territorial organizations and bodies
20	Ireland	Steam and hot water supply
21	Croatia	Extra-territorial organizations and bodies
22	Finland	Manufacture of tobacco products

23	Indonesia	Manufacture of coke oven products
24	Argentina	Extra-territorial organizations and bodies
25	Hungary	Extra-territorial organizations and bodies
26	Finland	Extraction of crude petroleum, gas and gas liquefaction
27	Finland	Extra-territorial organizations and bodies
28	China (except Hong Kong)	Extra-territorial organizations and bodies
29	Cyprus	Extraction of crude petroleum, gas and gas liquefaction
30	Cyprus	Extra-territorial organizations and bodies
31	Japan	Private households with employed persons
32	United States	Extra-territorial organizations and bodies
33	Luxembourg	Extraction of crude petroleum, gas and gas liquefaction
34	Luxembourg	Extra-territorial organizations and bodies
35	Malta	Manufacture of coke oven products
36	Czechia	Extra-territorial organizations and bodies
37	Romania	Extra-territorial organizations and bodies
38	Switzerland	Mining of coal and lignite, extraction of peat
39	Portugal	Manufacture of coke oven products
40	Canada	Steam and hot water supply
41	Australia	Private households with employed persons
42	Lithuania	Manufacture of coke oven products
43	Belgium	Steam and hot water supply
44	China (except Hong Kong)	Private households with employed persons
45	Denmark	Extra-territorial organizations and bodies
46	Malta	Manufacture of gas, distribution of gaseous fuels through mains
47	Norway	Extra-territorial organizations and bodies

48	Italy	Extra-territorial organizations and bodies
49	Türkiye	Extra-territorial organizations and bodies
50	Romania	Private households with employed persons
51	Mexico	Steam and hot water supply
52	South Africa	Extra-territorial organizations and bodies
53	Latvia	Extra-territorial organizations and bodies
54	Brazil	Extra-territorial organizations and bodies
55	Poland	Extra-territorial organizations and bodies
56	Malta	Steam and hot water supply
57	Australia	Extra-territorial organizations and bodies
58	Germany	Extra-territorial organizations and bodies
59	Netherlands	Extra-territorial organizations and bodies
60	Switzerland	Extra-territorial organizations and bodies
61	Saudi Arabia	Extra-territorial organizations and bodies
62	Brazil	Steam and hot water supply
63	Portugal	Mining of coal and lignite, extraction of peat
64	Cyprus	Steam and hot water supply
65	Luxembourg	Steam and hot water supply
66	Luxembourg	Petroleum Refinery
67	Ireland	Manufacture of tobacco products
68	France	Extra-territorial organizations and bodies
69	Luxembourg	Manufacture of coke oven products
70	Switzerland	Manufacture of coke oven products
71	Ireland	Extra-territorial organizations and bodies
72	Austria	Extra-territorial organizations and bodies
73	India	Extra-territorial organizations and bodies
74	Estonia	Manufacture of tobacco products
75	Luxembourg	Mining of Metals
76	Norway	Manufacture of coke oven products

77	Bulgaria	Extra-territorial organizations and bodies
78	Portugal	Extraction of crude petroleum, gas and gas liquefaction
79	Portugal	Extra-territorial organizations and bodies
80	Australia	Steam and hot water supply
81	Estonia	Extra-territorial organizations and bodies
82	Spain	Extra-territorial organizations and bodies
83	Lithuania	Extra-territorial organizations and bodies
84	Sweden	Manufacture of tobacco products
85	Canada	Extra-territorial organizations and bodies
86	Ireland	Manufacture of coke oven products
87	Belgium	Extra-territorial organizations and bodies

Table 6: List of "*problematic sectors*" aggregated with functionally similar sectors in the dataset preprocessing.

	Problematic sector	Aggregated together with sector	All
0	Private households with employed persons	Other service activities	Al countries
1	Manufacture of coke oven products	Petroleum Refinery	Latvia
2	Mining of coal and lignite, extraction of peat	Extraction of crude petroleum, gas and gas liquefaction	Malta
3	Research and development	Architectural and engineering activities, technical testing and analysis	Cyprus
4	Residential care activities and social work activities	Human health activities	Romania